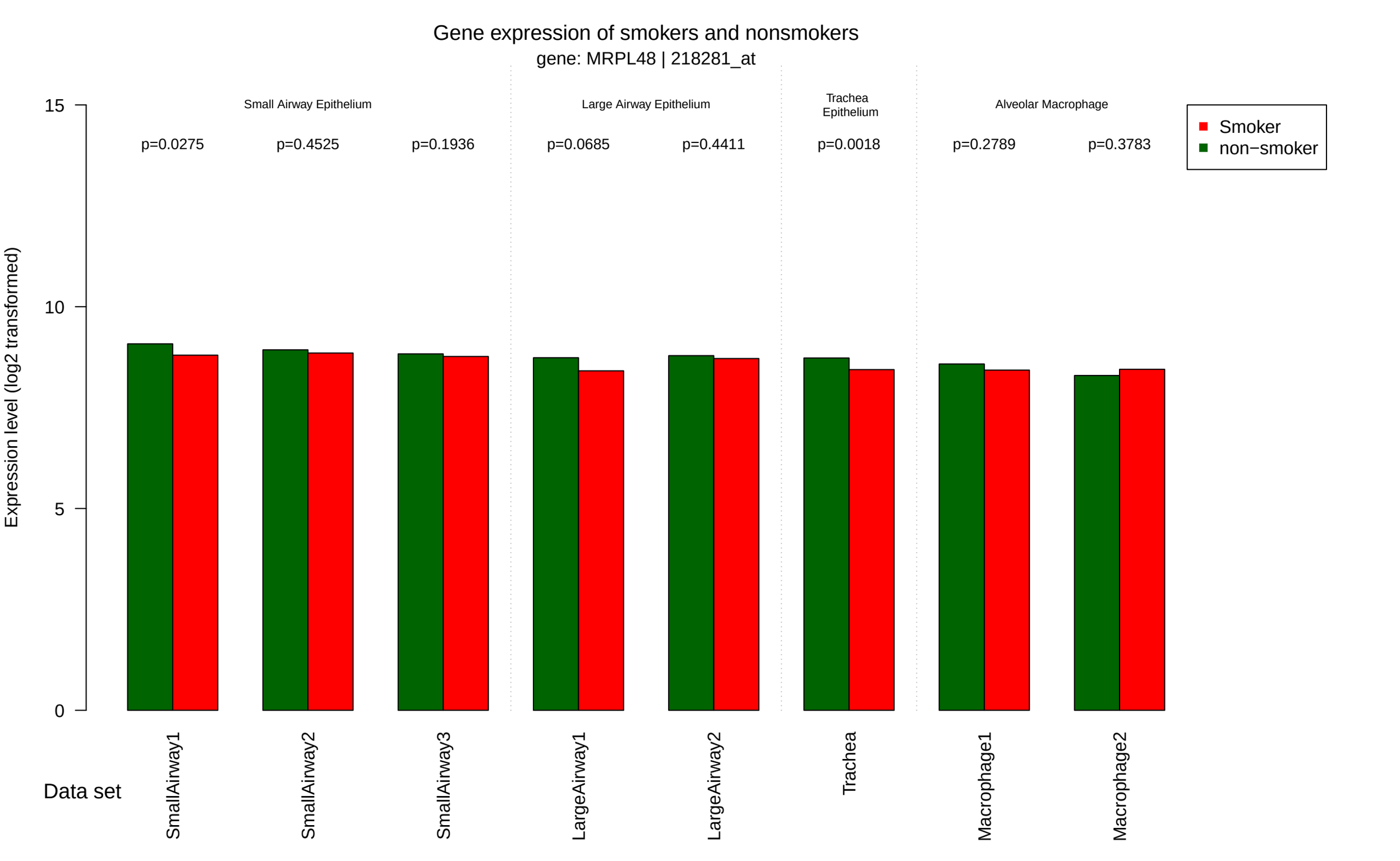
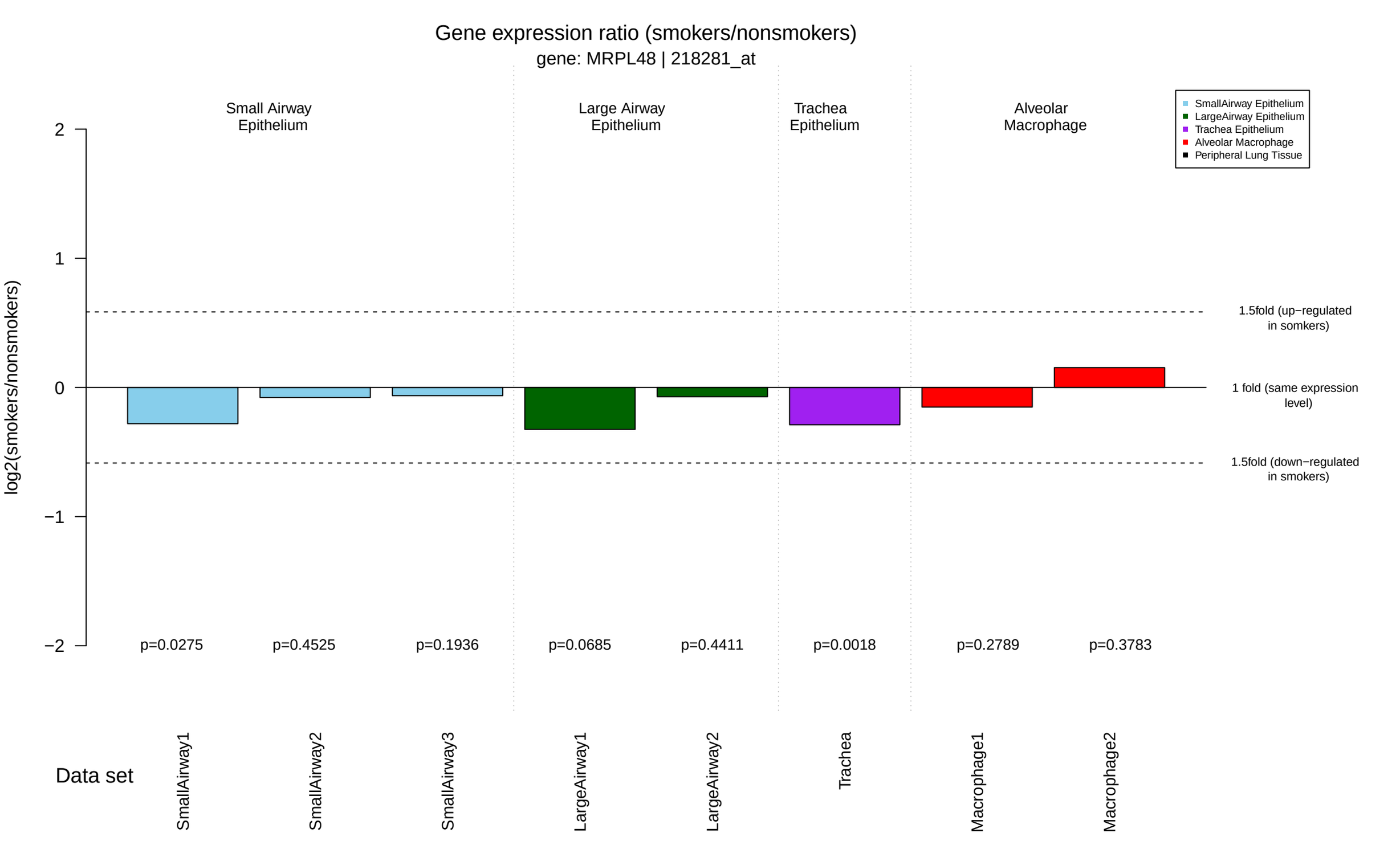
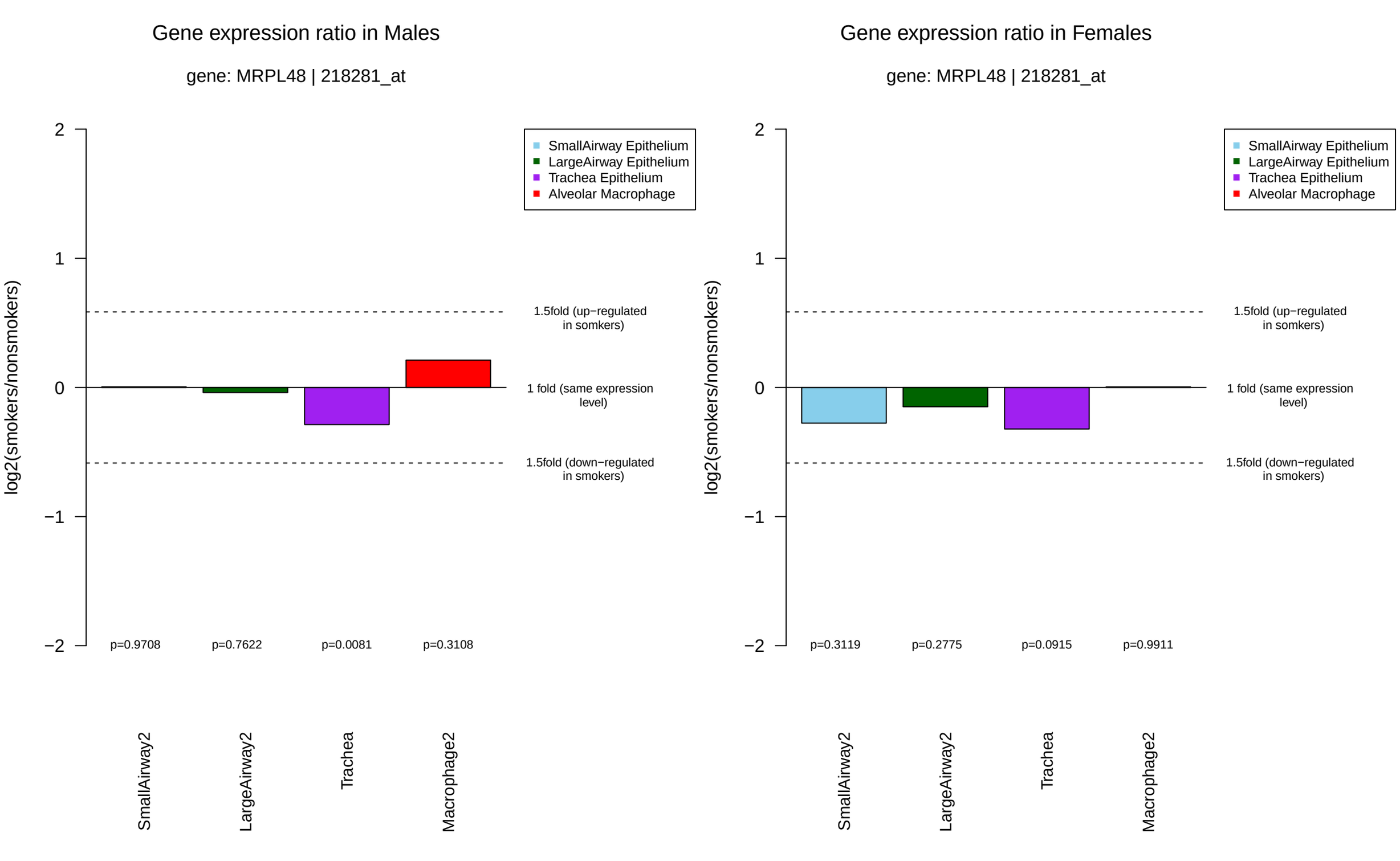
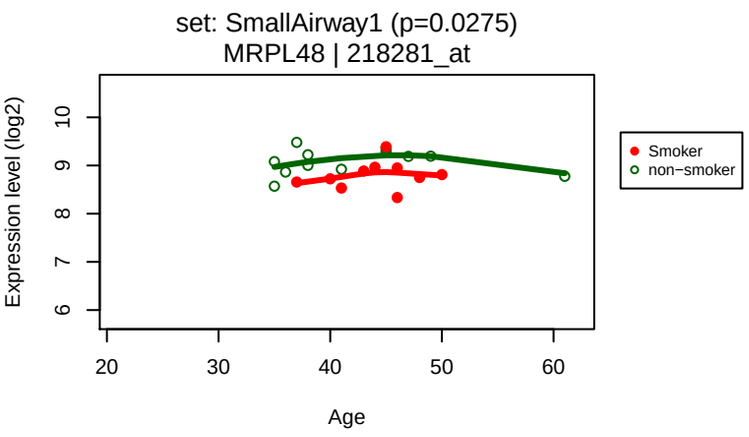
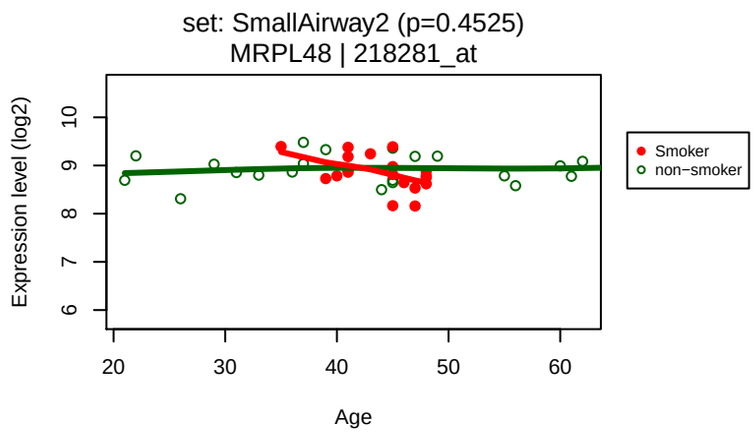
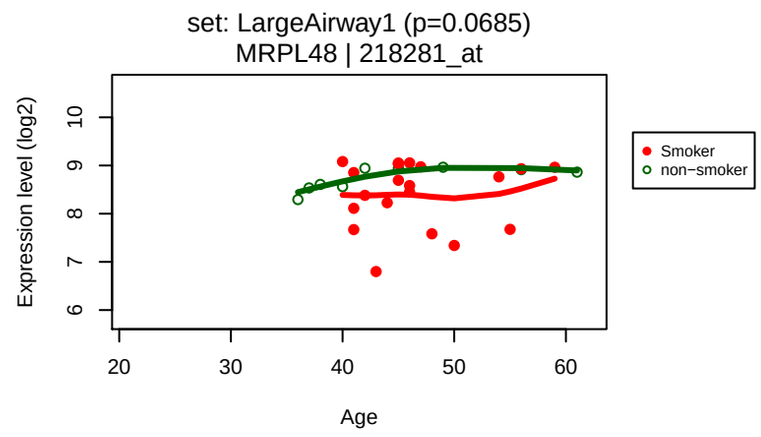

Gene expression of smokers and nonsmokers
gene: MRPL48 | 218281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0275
p=0.4525
p=0.1936
p=0.0685
p=0.4411
p=0.0018
p=0.2789
p=0.3783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

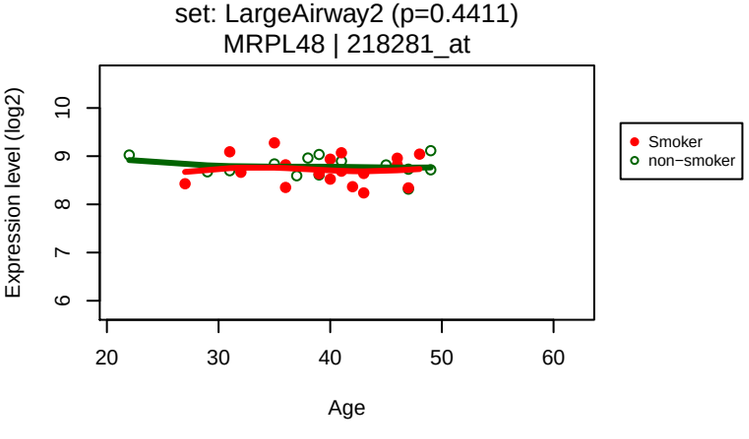

Gene expression ratio (smokers/nonsmokers)
gene: MRPL48 | 218281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0275
p=0.4525
p=0.1936
p=0.0685
p=0.4411
p=0.0018
p=0.2789
p=0.3783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

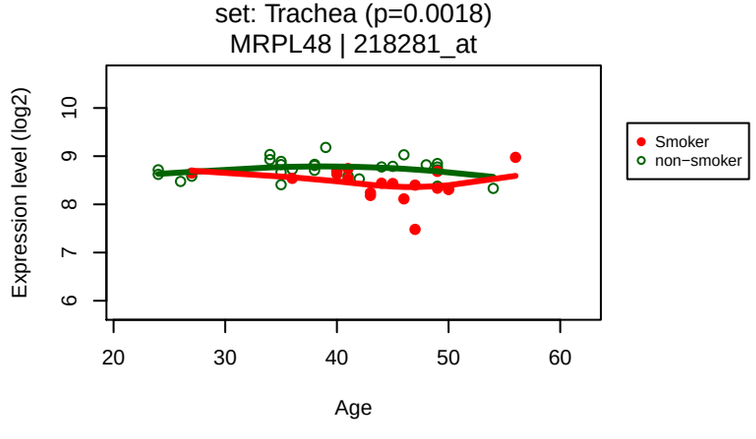

Gene expression ratio in Males
gene: MRPL48 | 218281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9708
p=0.7622
p=0.0081
p=0.3108
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL48 | 218281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3119
p=0.2775
p=0.0915
p=0.9911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

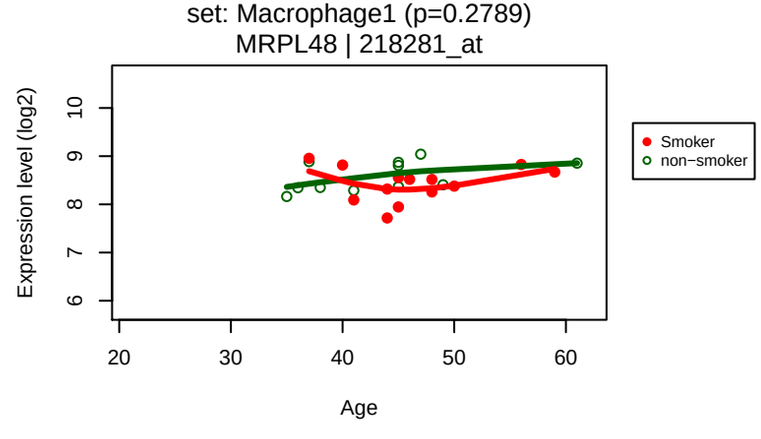

set: SmallAirway1 (p=0.0275)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker

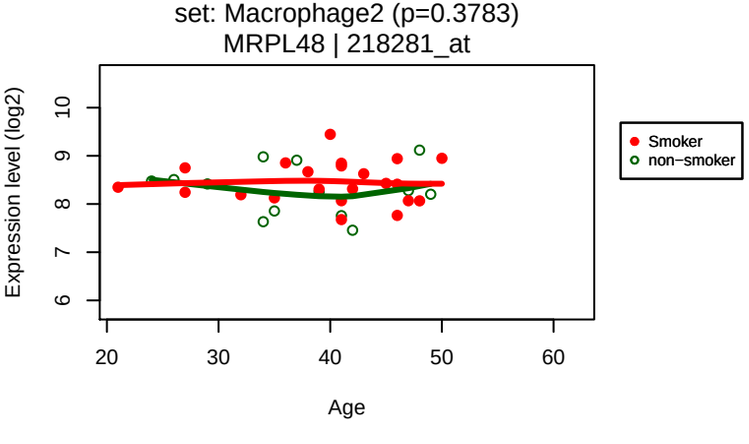

set: SmallAirway2 (p=0.4525)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker

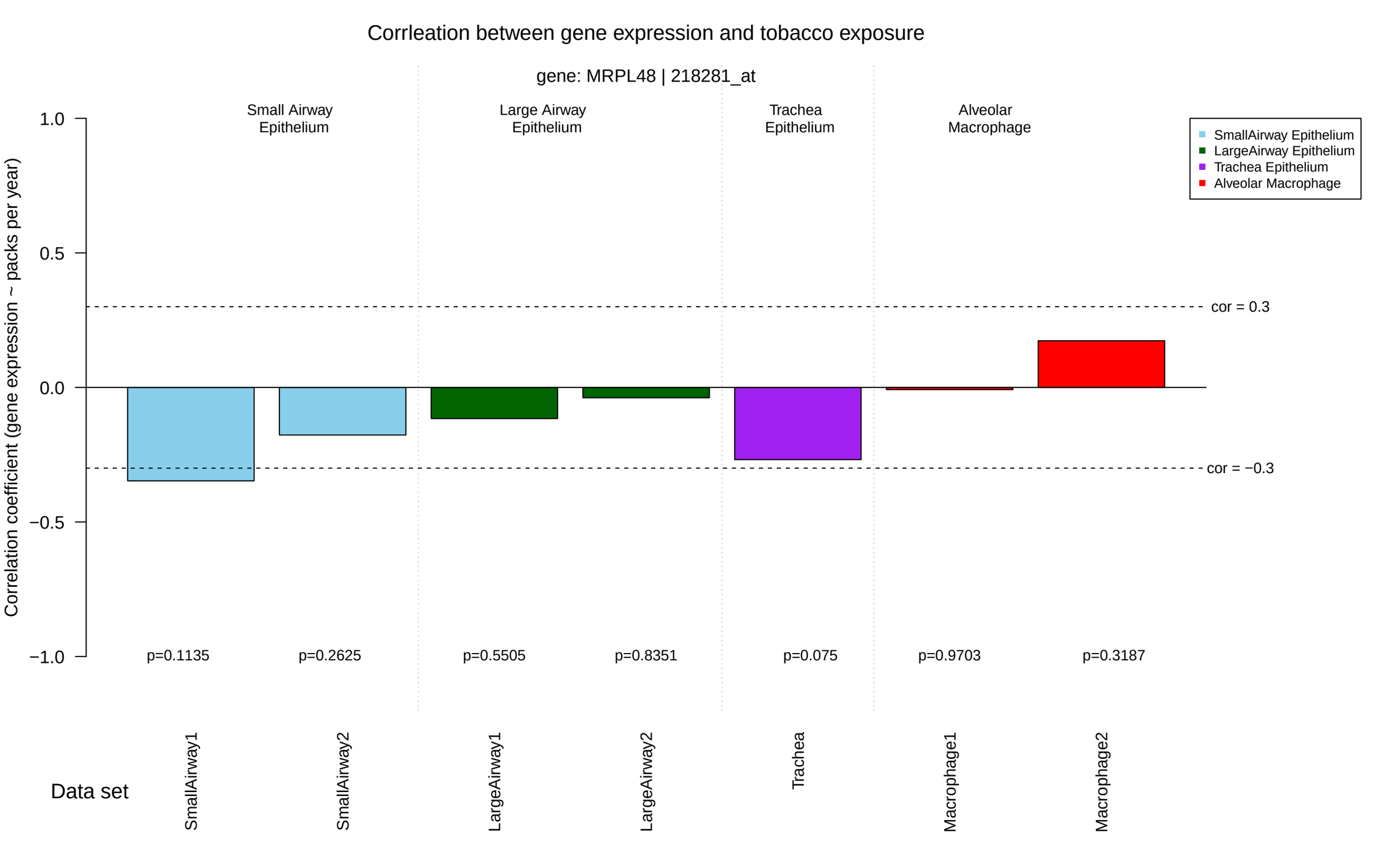

set: LargeAirway1 (p=0.0685)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4411)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2789)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3783)
MRPL48 | 218281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL48 | 218281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1135
p=0.2625
p=0.5505
p=0.8351
p=0.075
p=0.9703
p=0.3187
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2